W25 CMPUT 412/503 Exercise 1
Duckiebot Assembly and Basic Development

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Written report due: January 27th

Description

In this first exercise you will learn the basic knowledge and skills you need to operate a Duckiebot. This exercise will be marked **individually**, although we encourage you to work together with your labmates. It's to your benefit to read the whole lab, and keep an eye on the marking guide as you progress through the steps. Now, let's get quacking!
TIP: Read carefully. Most of the issues I encountered myself, and things I had a hard time finding, were because I was trying to rush through and skimming for commands to run.

Procedure

1) Set up your own course website
    a) This is where you will be hosting your written reports for each exercise
    b) You will be required to maintain your website all semester to provide a self-contained written report of your exercise solutions. It is also where you will host the video sections of your reports/demos.
    c) You can use whichever platform you prefer (e.g., Google site template or GitHub Pages with Jekyll)
    d) Written reports will be submitted as a pdf through eClass. The deadline is midnight the night before the first lab session for the next exercise. This means Tuesdays at 11:59 if you have lab on Wednesdays (LAB-H01-77989) and Thursdays at 11:59 if you have lab on Fridays (LAB-H02-77990), but please don't stay up later than you usually would working!

2) Set up your course GitHub repository
    a) You are expected to maintain a code repository on GitHub linked to your course webpage
    b) Include a README
    c) For each submission you will be asked to upload a link oneClass that points to the subfolder containing your code for the exercise

3) Read and do the following exercises in [RH1] Connecting and operating a Duckiebot. This manual is deprecated, but guidance on accomplishing the steps outlined can be found here.

a) Unit A-1 - Assembly duckumentation (**excluding** A-1.2)
    i) You will receive a pre-assembled Duckiebot
        (1) Be sure to carefully go over all the different components to test that they all work properly and you understand how they work. Once you are connecting to the duckiebot dashboard (later in the exercise), consider doing the hardware testing here
        (2) **Recommended**: go through the model DB21M build instructions; there is a troubleshooting section at the end that is useful
    ii) Take a picture of your Duckiebot and post it in the CMPUT412/503 W25 social-random channel. Also include photos of your build on your website report. Feel free to customize your rubber ducky placed on top!
    iii) **Warning**: the Duckiebot is equipped with a lithium-ion battery which can be dangerous if mishandled. Read this section to learn more about the Duckiebattery and how to handle it properly.
b) Unit A-2 - Terminal basics
c) Unit A-3 - Duckiebot Setup
    i) Your Duckiebot should already be initialized, please do not tinker with the SD card without consulting the TAs
    ii) This section is particularly bare in the old duckumentation. The "Software Setup" and "Duckiebot Operations" section of the current version of the manual has the steps you need
d) Unit A-4 - Networking basics
e) Unit A-5 - Docker basics
f) Unit A-6 - Basic Duckiebot operation

4) Read and do the following exercises in [RH2] Basic Development. Again, the further guidance on accomplishing the steps outlined can be found here.
    a) Unit B-1 - Git and GitHub
        i) You can skip the tutorial in sec. 1.1 of B-1 if you are already familiar with Git
    b) Unit B-2 - Python programs and environments
    c) Unit B-3 - Become a Docker Power-User
    d) Unit B-5 - Creating Docker containers (This is challenging and should be approached as a bonus)
    e) Unit B-6 - My First Duckietown Python Library

If you find yourself flush with time and want to use it to get a leg up on the next lab feel free to do the following! **THIS IS NOT FOR MARKS**.
Get your hands dirty with ROS:

f) Create a new package with `assignment1` name in your `ros_ws`, which will contain three nodes.
    i) First node will publish your **`NAME`** (string) to the topic **`name_listener`**
    ii) Second node will publish your **`STUDENT_ID`** (int) to the topic

`num_listener`

iii) Third node will subscribe to both the above topics and will print the following string: **`"quack quack! Duck cadet NAME checking in with ID STUDENT_ID!"`**, where NAME and STUDENT_ID will be replaced by what you're publishing.

iv) Publish using the first and the second node and subscribe to it from the third node 🦆

Deliverables

Include in your written report:

- A video of your Duckiebot driving in a straight line for a distance of 2 meters
- A video of your Duckiebot running the lane following demo
- A screen capture of the camera output and motor signals as seen from the Dashboard
- A screen capture of your intrinsics calibration.yaml file
- A screen capture of your extrinsics calibration.yaml file
- A screen capture of your Duckiebot saying “Hello from MY_ROBOT”
- Each of the above should include a short write up explaining the video or image.
- A short overall write up about what you implemented, how it works, what you learned, what challenges you came across, and how you overcame the challenges
- Your reflections, thoughts, and comments on this exercise of the lab

Questions to think about to help guide your write up:

- What is the purpose of calibrating the camera and motors?
- Why is docker useful?
- Say your robot started to veer right when driving in a straight line. What do you think is causing the drift? What could your robot do to detect and adjust for this drift?
- While running the lane following demo, at which parts of the road did your robot have trouble following the lane? Why do you think this is?
- How does your DuckieBot follow the lane in the lane following demo? If you don't know - can you speculate on how it works?
- What is your key takeaway from this lab?

On eClass you will submit:

- To submit your website written report, first publish it to the public then upload a pdf printout of your published report on eClass by the deadline
- A link to your new course website
- A link to your new course Github repository with all the code from this exercise neatly in a subfolder

Resources

You can use any material on the internet as long as you cite it properly. This includes tools like chatGPT. You are encouraged to collaborate with your labmates and if you develop a solution together please acknowledge who you worked with in your written report.

Marking Guide

Objective		Points
Website		/1
Set Up Repo (Git)		/1
	Setting up	/0.5
	Readme	/0.5
Report		/10
	Image of camera and motor signals from dashboard	/0.5
	Explanation of image	/0.5
	Screenshot of Duckiebot saying “Hello from MY_ROBOT” (RH2 Unit B-2.3)	/0.5
	Explanation of image	/0.5
	Screenshots of both camera calibration .yaml files (intrinsics and extrinsics)	/0.5
	Explanation of the images	/0.5
	Screenshot of kinematics .yaml file	/0.5
	Explanation of image	/0.5
	Video of 2m straight driving	/0.5
	Explanation of video	/0.5
	Video of lane following demo	/0.5
	Explanation of video	/0.5
	Write-up	/2
	Spelling and Grammar	/1
	References	/1

	Colour detector (bonus)	/up to +1
	What didn’t work (bonus)*	/up to 0.5 per incomplete robot based objective above
Total		**/12**

*If there is any objective you cannot complete, you have another way. In the “What I Learned” portion include a subsection for any objective or demonstration you cannot complete and include answers the following questions:

What was the barrier to completion?
What was the cause of that barrier?
What would you do next if you had more time?
Why is that what you would do next?
Anything else you learned while approaching this challenge.

Note that these bonus marks are unlikely to be given if the barrier is simply “I needed more time”. They are intended to allow you to explore problems and errors, come to an understanding, and share that learning with us to earn marks you lost because those errors prevented you from completing an objective.